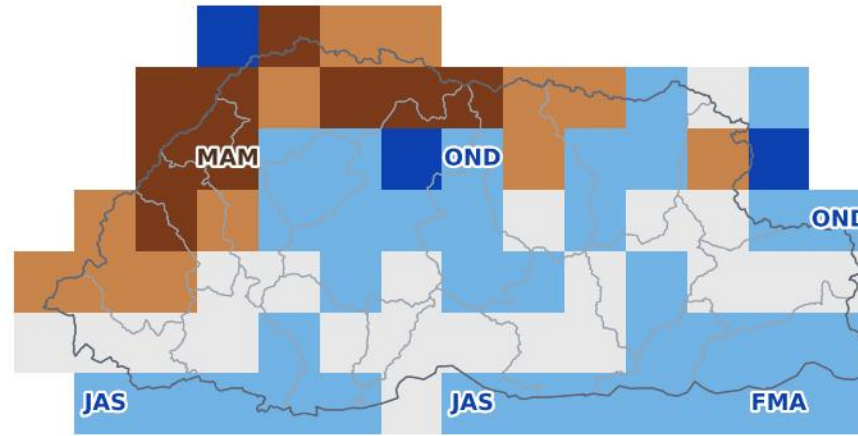


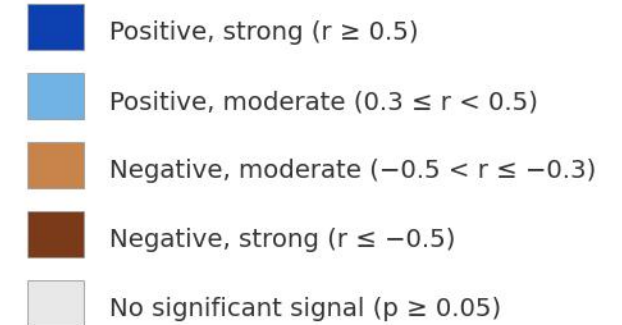
Bhutan: how does El Niño generally affect seasonal rainfall?

Pixelwise partial correlation of rainfall with Niño3.4 (ENSO), other climate modes held constant — ERA5 0.25° grid, 1981–2025

Unique ENSO signal — strongest significant trimester per pixel



Partial correlation with Niño3.4



Brown = El Niño years tend to be drier than normal in that season; blue = wetter.

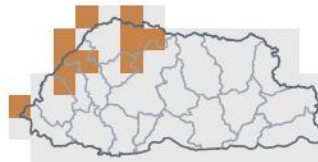
Method (the teleconnections page's partial pass):

- Trimester rainfall means per pixel, all 12 overlapping 3-month windows, 1981–2025
- Partial r of rainfall vs Niño3.4, holding DMI (IOD), TNA, TSA and AMM constant
- Each mode at its per-pixel best lag (0–3 mo)
- Two-tailed $p < 0.05$, df adjusted for controls
- All seasons kept (no rainy-season filter)

JFM



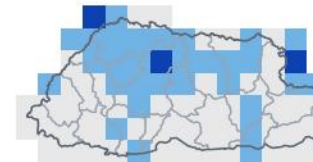
AMJ



JAS



OND

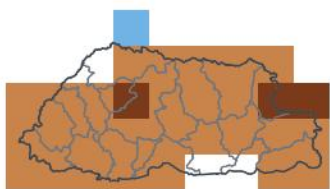


Data: ERA5 monthly precip (0.25°), NOAA PSL Niño3.4. Bottom row: the four quarters of the year.

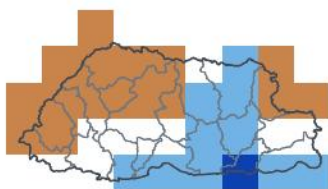
Bhutan: what seasonal rainfall is predicted this year (August-issued forecast)?

SEAS5 (issued August 2026) accounts for El Niño and every other driver · return-period categories vs the 1981–2025 hindcast · skill-shaded · ADM1

JJA 2026 — in season
1-in-5 dry · 18th pct



JAS 2026 — in season
1-in-2 dry · 44th pct



ASO 2026 — in season
1-in-2 wet · 51st pct



SON 2026 — lead 1 mo
1-in-2 dry · 40th pct



Forecast (high skill):

- Drought — ≥ 10 -yr return period
- Drought — 3–10-yr return period
- Roughly normal
- Flood — 3–10-yr return period
- Flood — ≥ 10 -yr return period

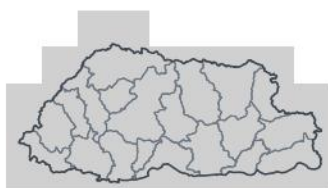
Other:

- Mod. skill
- Low skill
- Outside rainy season

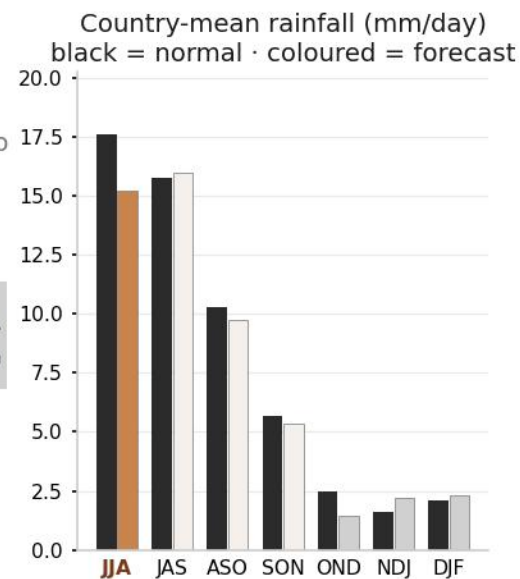
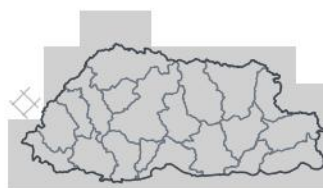
OND 2026 — lead 2 mo
off-season



NDJ 2026/27 — lead 3 mo
off-season



DJF 2026/27 — lead 4 mo
off-season



White hatch on a colour = moderate skill.
Tile subtitles: the country-level return period and percentile from the alerts app.
Driest outlook: JJA 2026 (1-in-5 dry).
In season = observed months + forecast.
Niño3.4: +1.7 °C in Jul 2026.